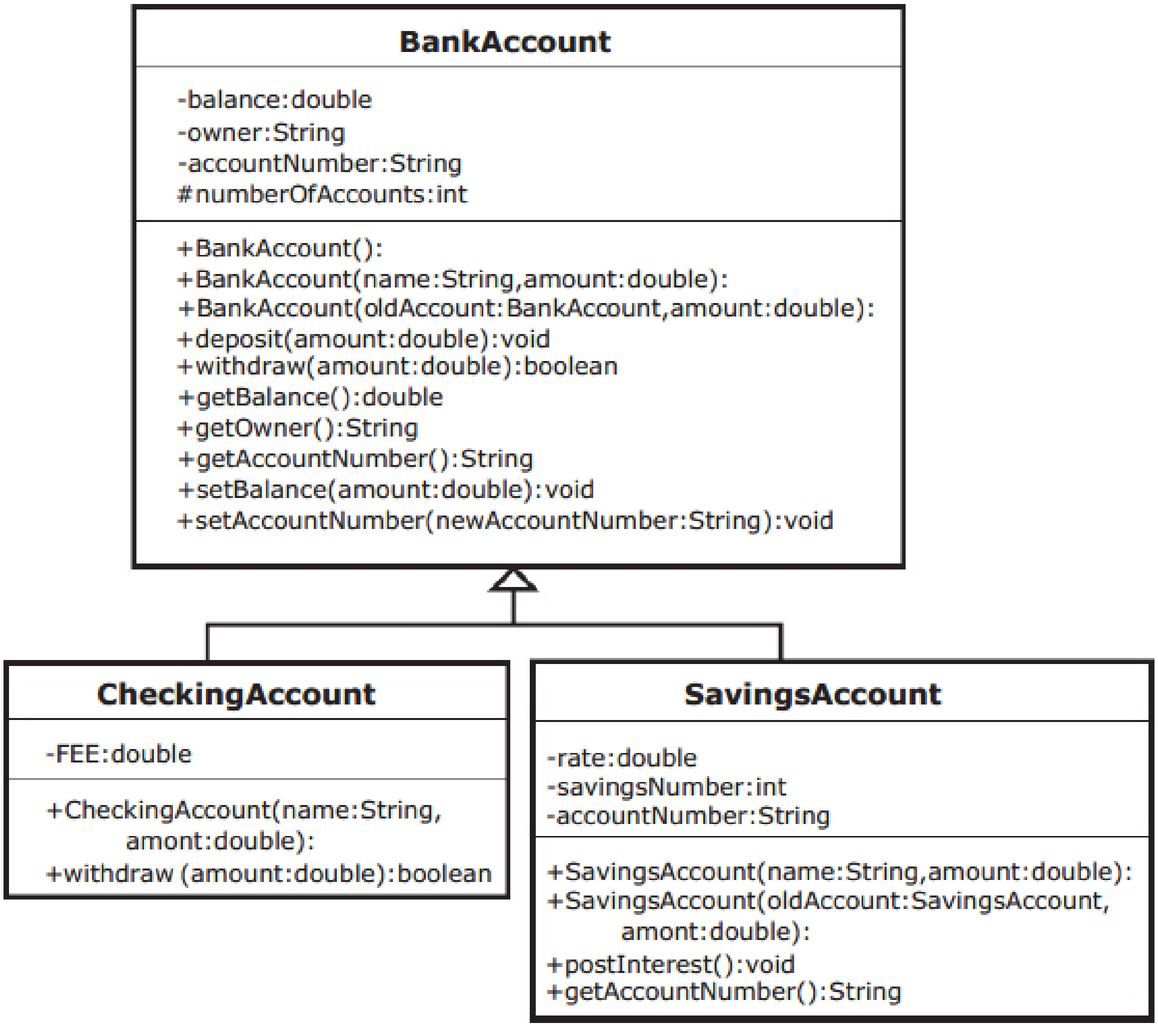
The UML diagram for the inheritance relationship is as follows:



Task #1 : Extending BankAccount

1. Copy the files AccountDriver.java (code listing 10.1) and *BankAccount.java* (code listing 10.2) BankAccount.java is complete and will not need to be modified.

2. Create a new class called **CheckingAccount** that extends **BankAccount**.

3. It should contain a static constant **FEE** that represents the cost of clearing one check. Set it equal to 15 cents.
4. Write a constructor that takes a name and an initial amount as parameters. It should call the constructor for the superclass. It should initialize **accountNumber** to be the

current value in **accountNumber** concatenated with –10 (All checking accounts at this bank are identified by the extension –10). There can be only one checking account for each account number.
Remember since accountNumber is a private member in **BankAccount**, it must be changed through a mutator method.

5. Write a new instance method, **withdraw**, that <u>overrides</u> the withdraw method in the superclass. This method should take the amount to withdraw, add to it the fee for check clearing, and call the withdraw method from the superclass.
   Remember that to override the method, it must have the same method heading. Notice that the withdraw method from the superclass returns true or false depending if it was able to complete the withdrawal or not. The method that overrides it must also return the same true or false that was returned from the call to the withdraw method from the superclass.

6. Compile and debug this class.

Task #2 Creating a Second Subclass

1. Create a new class called **SavingsAccount** that extends **BankAccount**.

2. It should contain an instance variable called rate that represents the annual interest rate. Set it equal to 2.5%.

3. It should also have an instance variable called **savingsNumber**, initialized to 0. In this bank, you have one account number, but can have several savings accounts with that same number. Each individual savings account is identified by the number following a dash. For example, 100001-0 is the first savings account you open, 100001-1 would be another savings account that is still part of your same account. This is so that you can keep some funds separate from the others, like a Christmas club account.

4. An instance variable called **accountNumber** that will hide the **accountNumber** from the superclass, should also be in this class.

5. Write a constructor that takes a name and an initial balance as parameters and calls the constructor for the superclass. It should initialize **accountNumber** to be the current value in the superclass **accountNumber** (the hidden instance variable) concatenated with a hyphen and then the **savingsNumber**.

6. Write a method called **postInterest** that has no parameters and returns no value. This method will calculate one month's worth of interest on the balance and deposit it into the account.

7. Write a method that overrides the **getAccountNumber** method in the superclass.
8. Write a copy constructor that creates another savings account for the same person. It should take the original savings account and an initial balance as parameters. It should call the copy constructor of the superclass, assign the **savingsNumber** to be one more than the **savingsNumber** of the original savings account. It should assign the **accountNumber** to be the **accountNumber** of the superclass concatenated with the hypen and the **savingsNumber** of the new account.

9. Compile and debug this class.

10. Use the **AccountDriver** class to test out your classes. If you named and created your classes and methods correctly, it should not have any difficulties. If you have errors, do not edit the **AccountDriver** class. You must make your classes work with this program.

11. Running the program should give the following output:

```
Account Number 100001-10 belonging to Benjamin Franklin
Initial balance = $1000.00
After deposit of $500.00, balance = $1500.00
After withdrawal of $1000.00, balance = $499.85
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Control Structures to Objects
Account Number 100002-0 belonging to William Shakespeare
Initial balance = $400.00
After deposit of $500.00, balance = $900.00
Insuffient funds to withdraw $1000.00, balance = $900.00
After monthly interest has been posted, balance = $901.88

Account Number 100002-1 belonging to William Shakespeare
Initial balance = $5.00
After deposit of $500.00, balance = $505.00
Insuffient funds to withdraw $1000.00, balance = $505.00

Account Number 100003-10 belonging to Isaac Newton
```

Code listing 10.1

```
//Demonstrates the BankAccount and derived classes

import java.text.*;              // to use Decimal Format

public class AccountDriver
{
     public static void main(String[] args)
```

```
    {
        double put_in = 500;
        double take_out = 1000;

        DecimalFormat myFormat;
        String money;
        String money_in;
        String money_out;
        boolean completed;

        // to get 2 decimals every time
        myFormat = new DecimalFormat("#.00");

        //to test the Checking Account class
        CheckingAccount myCheckingAccount =
                new CheckingAccount ("Benjamin Franklin", 1000);
        System.out.println ("Account Number "
             + myCheckingAccount.getAccountNumber() +
             " belonging to " + myCheckingAccount.getOwner());
        money  = myFormat.format(myCheckingAccount.getBalance());
        System.out.println ("Initial balance = $" + money);
        myCheckingAccount.deposit (put_in);
        money_in = myFormat.format(put_in);
        money  = myFormat.format(myCheckingAccount.getBalance());
        System.out.println ("After deposit of $" + money_in
             + ",  balance = $" + money);
        completed = myCheckingAccount.withdraw(take_out);
        money_out = myFormat.format(take_out);
        money  = myFormat.format(myCheckingAccount.getBalance());
        if (completed)
        {
             System.out.println ("After withdrawal of $" +
money_out
                        + ",  balance = $" + money);
        }
        else
        {
             System.out.println ("Insuffient funds to withdraw $"
                        + money_out     + ",  balance = $" +
money);
        }
        System.out.println();

        //to test the savings account class
        SavingsAccount yourAccount =
                new SavingsAccount ("William Shakespeare", 400);
        System.out.println ("Account Number "
                + yourAccount.getAccountNumber() +
                " belonging to " + yourAccount.getOwner());
        money  = myFormat.format(yourAccount.getBalance());
        System.out.println ("Initial balance = $" + money);
        yourAccount.deposit (put_in);
```

```
          money_in = myFormat.format(put_in);
          money  = myFormat.format(yourAccount.getBalance());
          System.out.println ("After deposit of $" + money_in
                     + ",  balance = $" + money);
          completed = yourAccount.withdraw(take_out);
          money_out = myFormat.format(take_out);
          money  = myFormat.format(yourAccount.getBalance());
          if (completed)
          {
               System.out.println ("After withdrawal of $" +
money_out
                     + ",  balance = $" + money);
          }
          else
          {
               System.out.println ("Insuffient funds to withdraw $"
                     + money_out     + ",  balance = $" + money);
          }
          yourAccount.postInterest();
          money  = myFormat.format(yourAccount.getBalance());
          System.out.println ("After monthly interest has been
posted,"
                     + "balance = $"  + money);
          System.out.println();


          // to test the copy constructor of the savings account
class
          SavingsAccount secondAccount =
                     new SavingsAccount (yourAccount,5);
          System.out.println ("Account Number "
                     + secondAccount.getAccountNumber()+
                     " belonging to " + secondAccount.getOwner());
          money  = myFormat.format(secondAccount.getBalance());
          System.out.println ("Initial balance = $" + money);
          secondAccount.deposit (put_in);
          money_in = myFormat.format(put_in);
          money  = myFormat.format(secondAccount.getBalance());
          System.out.println ("After deposit of $" + money_in
                     + ", balance = $" + money);
          secondAccount.withdraw(take_out);
          money_out = myFormat.format(take_out);
          money  = myFormat.format(secondAccount.getBalance());
          if (completed)
          {
               System.out.println ("After withdrawal of $" +
money_out
                          + ",  balance = $" + money);
          }
          else
          {
               System.out.println ("Insuffient funds to withdraw $"
```

```
                            + money_out     + ",  balance = $" +
money);
           }
           System.out.println();

           //to test to make sure new accounts are numbered correctly
           CheckingAccount yourCheckingAccount =
                      new CheckingAccount ("Issac Newton", 5000);
           System.out.println ("Account Number "
                      + yourCheckingAccount.getAccountNumber()
                      + " belonging to "
                      + yourCheckingAccount.getOwner());

      }
}
```

Code listing 10.2

```
//Defines any type of bank account

public abstract class BankAccount
{
      // class variable so that each account has a unique number
      protected static int numberOfAccounts = 100001;

      // current balance in the account
      private double balance;
      // name on the account
      private String owner;
      // number bank uses to identify account
      private String accountNumber;

      //default constructor
      public BankAccount()
      {
           balance = 0;
           accountNumber = numberOfAccounts + "";
           numberOfAccounts++;
      }

      //standard constructor
      public BankAccount(String name, double amount)
      {
           owner = name;
           balance = amount;
           accountNumber = numberOfAccounts + "";
           numberOfAccounts++;
      }

      //copy constructor
```

```
public BankAccount(BankAccount oldAccount, double amount)
{
      owner = oldAccount.owner;
      balance = amount;
      accountNumber = oldAccount.accountNumber;
}

//allows you to add money to the account
public void deposit(double amount)
{
      balance = balance + amount;
}

//allows you to remove money from the account if
//enough money is available
//returns true if the transaction was completed
//returns false if the there was not enough money
public boolean withdraw(double amount)
{
      boolean completed = true;

      if (amount <= balance)
      {
            balance = balance - amount;
      }
      else
      {
            completed = false;
      }
      return completed;
}

//accessor method to balance
public double getBalance()
{
      return balance;
}

//accessor method to owner
public String getOwner()
{
      return owner;
}

//accessor method to account number
public String getAccountNumber()
{
      return accountNumber;
}

//mutator method to change the balance
public void setBalance(double newBalance)
```

```
    {
        balance = newBalance;
    }

    //mutator method to change the account number
    public void setAccountNumber(String newAccountNumber)
    {
        accountNumber = newAccountNumber;
    }
}
```